

combination of microcontrollers and microprocessors will deliver the best performance and efficiency.

As for the physical design of an electronic product, it makes sense to work with a partner who is highly experienced in PCB layout and schematic design. Product designers need to keep several things in mind like types of design tools, software for product design and hardware, among others.

Prototype, test, analyse and redefine

To overcome the barriers to market entry, a third party might help in making speedy but necessary design iterations. Before a product enters the market, it must be prototyped, tested, analysed and refined, and refined again. Along with testing and verification, manufacturers need to get their newly designed products registered under Compulsory Regulatory Scheme (CRS). This has been introduced by Ministry of Electronics and Information Technology (MeitY), and with Bureau of Indian Standards (BIS).

Once a perfect market fit and adequate funding is found for the product, having a third-party engineering team with extensive experience in designing and helping to bring products to market can help significantly in making the most of funding (from sources like Ketto, an Indian online crowd-sourcing platform) and moving the whole process forward.

A partner to help execute

The journey from concept to customer is lengthy and full of hurdles. One of the biggest challenges in this process is executing the product idea. If anything goes wrong at this stage, a lot of hard work, time and money might get wasted. That is why it is essential for the product creator to have an excellent relationship with a strong hardware partner, who has the expertise to manufacture the finished item in a way that guarantees optimum quality, keeping production costs to an absolute minimum. Vital elements in manufacturing include sourcing the most suitable materials and components with bill of materials cost optimisation.

Dragon Innovation, for example, is a US-based product development firm with offices in the US and China. It supports product creators to move from design to prototype to volume production. The company helps its customers understand manufacturing costs and schedules, find factories and manage production as they scale their businesses.

In particular, organisations that offer product development support can enable their customers to gain insights on cash flow, break-even reports and systems of record-keeping, throughout the manufacturing journey.

Marketing and distribution

The next stage is high-service distribution. This involves kitting, box design and build, printing of manuals and user guides, accessory bundling, and individual and bulk packaging.

During this stage, the distributor needs to have experience in handling vast quantities of electronic products efficiently and cost-effectively, especially online, where access to a massive engineering community is available.

Finally, the product needs to be marketed to potential customers. This stage could include an official product launch through printed and online media—particularly, vitally important social media outlets—that target the product's intended market.

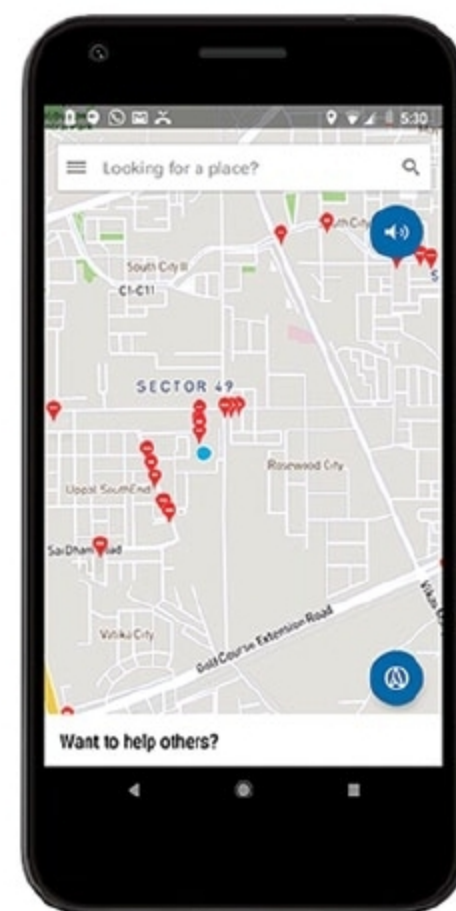
For instance, Premier Farnell has supported the global technology community at every stage of the product development process, helping design engineers create designs, build prototypes and deliver finished products quickly. The company has recently launched a new service for makers to help them bring their product ideas to market. At the same time, its distribution channels ensure products reach the largest markets possible.

The overall goal is always to help people, products and businesses to apply technology that will change the world for the better.

—Ravi Pagar, regional director - South Asia and ASEAN, element14

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An Automated Navigation Technology To Reduce Road Accidents



Intents app

Do you know that more than 11,000 road accidents are caused in our country every year due to potholes and speed breakers? Can this be solved using advanced navigation technology with smartphone as the primary device?

Tabrez Alam, Bala Subramaniam and Jay Prakash formed Pluggdd Mobile in 2017, keeping this in mind. Alam says, "One of the most common devices used by people is a cellphone. Hence, we created Intents app, whose technology is entirely focussed on using the sensors of a cellphone. It is a complete navigation app, which also alerts users about speed breakers, potholes and the like on the road ahead."

The app uses two basic cellphone sensors: accelerometer and GPS. Since